

Lab: Subnet Masking and Host Calculation (IPv4 and IPv6)

Objective

To gain hands-on experience with subnet masking, subnet calculation, and host calculation for IPv4 and IPv6 addresses.

Task 1: IPv4 Subnet Mask Calculation

Instructions

You have an IP address with a specific subnet mask. Your task is to determine the subnet mask in decimal format.

Given

- IP Address: 192.168.1.0/24

Solution

1. Determine the subnet mask

- The /24 notation indicates that the first 24 bits of the IP address are of the network portion of the address.
- This results in the subnet mask 255.255.255.0 in decimal format. Each octet of the subnet mask represents eight bits set to 1 (255) or 0 (0), corresponding to the /24 prefix.

2. Verifying the answer through an online subnet masking calculator

- In this lab, you will verify the calculation using Visual Subnet Calculator <https://www.davidec.net/sites/default/subnets/subnets.html>. (Note: There are many IP and Netmask Calculators available online. You can use any).
- Open the Visual Subnet Calculator
- Enter Network Address 192.168.1.0 and Mask Bit 24
- Select the option **Netmask** in **Show columns**:
- Click **Update** and check that the subnet mask you calculated matches the result shown by the calculator.

Visual Subnet Calculator

Enter the network you wish to subnet:

Network Address

Mask bits

192.168.1.0

/24

Update

Reset

Show columns: ☒ Subnet address ☒ Netmask ☒ Range of addresses ☒ Useable IPs ☒ Hosts ☒ Divide ☒ Join

Click below to split and join subnets.
If you wish to save this subnetting for later, bookmark [this hyperlink](#).

Subnet address	Netmask	Range of addresses	Useable IPs	Hosts	Divide	Join
192.168.1.0/24	255.255.255.0	192.168.1.0 - 192.168.1.255	192.168.1.1 - 192.168.1.254	254	Divide	/24

Task 2: IPv4 subnet calculation

Instructions

You need to divide a given network into smaller subnets. Calculate the new subnet mask and list the resulting subnets with their address ranges.

Given

- Network address: 10.0.0.0/24
- Required subnets: 4

Solution

1. Calculate the new subnet mask:

- To create 4 subnets, you need to borrow 2 bits from the host portion of the IP address.
- This changes the CIDR notation to /26 (24 original network bits + 2 borrowed bits).
- The new subnet mask is 255.255.255.192.

2. List subnets:

- Subnet 1:** 10.0.0.0/26 with the range 10.0.0.0 - 10.0.0.63
- Subnet 2:** 10.0.0.64/26 with the range 10.0.0.64 - 10.0.0.127
- Subnet 3:** 10.0.0.128/26 with the range 10.0.0.128 - 10.0.0.191
- Subnet 4:** 10.0.0.192/26 with the range 10.0.0.192 - 10.0.0.255

3. Verifying the answer through an online subnet masking calculator

Subnet 1

- Open the Visual Subnet Calculator: (<https://www.davide.net/sites/default/subnets/subnets.html>)
- Check the Subnet 1.
- Enter Network Address 10.0.0.0 and Mask Bit 26 as shown in the calculation for Subnet 1.
- Verify that the Subnet address, Netmask, and Range of addresses are selected in **Show columns**.
- Click **Update** and check the calculator output.
- Observe the Subnet address, Netmask, Range of addresses, and other information r shown in the online calculator.

Visual Subnet Calculator

Enter the network you wish to subnet:

Network Address

Mask bits

10.0.0.0

/26

Update

Reset

Show columns: ☒ Subnet address ☒ Netmask ☒ Range of addresses ☒ Useable IPs ☒ Hosts ☒ Divide ☒ Join

Click below to split and join subnets.
If you wish to save this subnetting for later, bookmark [this hyperlink](#).

Subnet address	Netmask	Range of addresses	Useable IPs	Hosts	Divide	Join
10.0.0.0/26	255.255.255.192	10.0.0.0 - 10.0.0.63	10.0.0.1 - 10.0.0.62	62	Divide	/26

Subnet 2

- Similarly, let us check the calculation for Subnet1.
- Enter Network Address 10.0.0.64 and Mask Bit 26 as shown in the calculation for Subnet 2.
- Make sure the option Subnet address, Netmask, and Range of addresses are selected in **Show columns**.
- Click **Update** and select the calculator output.
- Observe the Subnet address, Netmask, Range of addresses, Usable IPs, and others options displayed in the online calculator.

Visual Subnet Calculator

Enter the network you wish to subnet:

Network Address

Mask bits

10.0.0.64

/26

Update

Reset

Show columns: ☒ Subnet address ☒ Netmask ☒ Range of addresses ☒ Useable IPs ☒ Hosts ☒ Divide ☒ Join

Click below to split and join subnets.
If you wish to save this subnetting for later, bookmark [this hyperlink](#).

Subnet address	Netmask	Range of addresses	Useable IPs	Hosts	Divide	Join
10.0.0.64/26	255.255.255.192	10.0.0.64 - 10.0.0.127	10.0.0.65 - 10.0.0.126	62	Divide	/26

- You can address the calculation for Subnet 3 and Subnet 4, which you can verify similarly.

Task 3: IPv4 host calculation

Instructions

Given a subnet mask, calculate the number of usable hosts within that subnet.

Given

- Network address: 10.10.0.0
- Subnet mask: 255.255.255.192

Solution

1. Calculate the number of hosts:

- The subnet mask 255.255.255.192 corresponds to a /26 prefix, meaning there are 6 bits available for host addresses (32 total bits minus 26 network bits).
- The number of usable hosts is calculated as $2^6 - 2 = 62$ (subtracting 2 accounts for the network and broadcast addresses).

2. Verifying the answer through an online subnet masking calculator

- Open the Visual Subnet Calculator (<https://www.davide.net/sites/default/subnets/subnets.html>)
- Enter Network Address 10.10.0.0 and Mask Bit 26.
- Make sure the options Subnet address, Netmask, Range of addresses, Usable IPs, and Hosts are selected in the Show columns.
- Click **Update** and check the calculator output.
- Observe the Subnet address, Netmask, Range of addresses, Usable IPs, and Hosts shown in the online calculator.

Visual Subnet Calculator

Enter the network you wish to subnet:

Network AddressMask bits

10.10.0.0/26

UpdateReset

Show columns: ☒ Subnet address ☒ Netmask ☒ Range of addresses ☒ Useable IPs ☒ Hosts ☒ Divide ☒ Join

Click below to split and join subnets.
If you wish to save this subnetting for later, bookmark [this hyperlink](#).

Subnet address	Netmask	Range of addresses	Useable IPs	Hosts	Divide	Join
10.10.0.0/26	255.255.255.192	10.10.0.0 - 10.10.0.63	10.10.0.1 - 10.10.0.62	62	Divide	Join

Task 4: IPv6 subnetting

Instructions

You are given an IPv6 address and need to create subnets. Calculate the new subnet mask and list a few of the resulting subnets.

Given

- IPv6 address: 2001::0db8:85a3::/48
- Required subnets: 16

Solution:

1. Calculate the new subnet mask:

- To create 16 subnets, you need to borrow 4 bits from the host portion of the IPv6 address.
- This changes the CIDR notation to /52 (48 original network bits + 4 borrowed bits).
- The new subnet mask is ffff::ffff:ffff:ffff:ffff:ffff:ffff:0000.

2. List a few subnets:

- Subnet 1: 2001::0db8:85a3::0000::/52
- Subnet 2: 2001::0db8:85a3::0010::/52
- Subnet 3: 2001::0db8:85a3::0020::/52
- Subnet 4: 2001::0db8:85a3::0030::/52

Task 5: IPv6 host calculation

Instructions

Given an IPv6 Address and subnet mask, calculate the possible hosts within that subnet.

Given

- IP Address: 2001::0db8:85a3::0000:0000:8a2e:0370:7334
- Subnet Mask: ffff::ffff:ffff:ffff:0000:0000:0000:0000 (CIDR /64)

Solution

1. Calculate the number of hosts:

- The subnet mask ffff::ffff:ffff:ffff:0000:0000:0000:0000 corresponds to a /64 prefix, meaning there are 64 bits available for host addresses (128 total bits minus 64 network bits).
- The number of possible hosts is 2^{64} , which equals approximately 1.84×10^{19} addresses.

2. Verifying the answer through an online subnet masking calculator

- Open the IPv6 Subnet Calculator (<https://www.vultr.com/resources/subnet-calculator-ipv6/>). (Note: There are many IPv6 and Netmasking Calculators available online. You can use any).
- Enter IP Address 2001::0db8:85a3::0000:0000:8a2e:0370:7334 and Mask Bit /64 i.e. 2001::0db8:85a3::0000:0000:8a2e:0370:7334/64
- Click **Calculate Subnet** and check the calculator output
- Observe usable addresses in the output.(Note: Not all the IP addresses are usable, as few are reserved.)

**Password Generator**
Generate randomized, strong, secure passwords.

**IPv4 Calculator**
Convert IPv4 CIDR notation.

**IPv6 Calculator**
Convert IPv6 CIDR notation.

**MAC Converter**
Convert a mac address between integer, hexadecimal, dot notation and more

**IPv4 Converter**
Convert IP addresses to decimal format, integer format, and more!

**Unixtime Calculator**
Convert between unix timestamp and datetime formats.

**IPv6 Subnet Calculator**
Convert IPv6 CIDR notation.

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Calculate Subnet

Network breakout

CIDR Notation ⓘ

2001:0db8:85a3:0000:0000:0000:0000/64

Address

2001:0db8:85a3:0000:0000:8a2e:0370:7334

Address Range Start

2001:0db8:85a3:0000:0000:0000:0000:0000

Address Range End

2001:0db8:85a3:0000:ffff:ffff:ffff:ffff

Mask Bits

64

Display

long

Usable Addresses

18,446,744,073,709,551,616

Available Subnets

/64 1 network

Exercises

IPv4 Exercises

Exercise 1: IPv4 Subnet Mask Calculation

Instructions

Given an IP address and a CIDR prefix, determine the subnet mask in decimal format.

Given

- IP Address: 172.16.0.0/20

Hint:

The /20 notation means the first 20 bits are for the network portion. Convert this to a decimal subnet mask.

Exercise 2: IPv4 Subnet Calculation

Instructions

You need to divide a given network into smaller subnets. Calculate the new subnet mask and list a few resulting subnets with their address ranges.

Given

- Network Address: 192.168.10.0/24
- Required Subnets: 8

Hint

To create 8 subnets, you'll need to borrow a certain number of bits from the host portion. Adjust the CIDR notation accordingly and determine the new subnet mask.

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IPv6 Exercise

Exercise 1: IPv6 Subnetting

Instructions

You are given an IPv6 address and need to create subnets. Calculate the new subnet mask and list a few of the resulting subnets.

Given

- IPv6 Address: 2001:abcd:1234::/48
- Required Subnets: 4

Hint

To create 4 subnets, you need to borrow 2 bits from the host portion. Adjust the CIDR notation and subnet mask accordingly.

Conclusion

You have gained hands-on experience in this lab with subnet masking and host calculations for IPv4 and IPv6 addresses. You learned how to determine the subnet mask in decimal format from CIDR notation, which is fundamental for understanding network segmentation. You practiced dividing a network into smaller subnets, calculating new subnet masks, and listing resulting subnets with their address ranges, which is crucial for efficient network design. Additionally, you explored IPv6 subnetting by calculating new subnet masks and listing subnets, emphasizing the importance of managing large address spaces effectively. These exercises enhance your skills in optimizing network performance, improving security, and managing address allocations in real-world scenarios.

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